4.1.2 Indexing a document for the first time

Let's find out what happens if we index a document without creating the schema up front. Suppose we want to index a `movie` document, as shown in the following listing.

Listing 4.1 Indexing a document with ID 1 into a `movies` index

```
PUT movies/_doc/1
{
  "title":"Godfather",
  "rating":4.9,
  "release_year":"1972/08/01"
}
```

Title of the movie

Rating given to the movie

Movie's release year (note the date format)

This is the first document sent to Elasticsearch to be indexed. Remember, we didn't create an index (`movies`) or schema for this document data before document ingestion. Here's what happens when this document hits the engine:

1. A new index (`movies`) is created automatically with default settings.
2. A new schema is created for the `movies` index using data types (discussed shortly) deduced from this document's fields. For example, `title` is set to the `text` and `keyword` types, `rating` to `float`, and `release_year` to a `date` type.
3. The document is indexed and stored in the Elasticsearch data store.
4. Subsequent documents are indexed without undergoing the previous steps because Elasticsearch consults the newly created schema for further indexing.

Elasticsearch does some groundwork behind the scenes when performing these steps to create a schema definition with field names and additional information. We can use the mapping API to dynamically fetch the schema definition that Elasticsearch created. The response to a `GET` command is shown in figure 4.2.

```
GET movies/_mapping
...
 "properties" : {
   "rating" : {"type" : "float"},
   "release_year" : {
     "type" : "date",
     "format" : "yyyy/MM/dd HH:mm:ss||yyyy/MM/dd||epoch_millis"
   },
   "title" : {
     "type" : "text",
     "fields" : {
       "keyword" : {
         "type" : "keyword",
         "ignore_above" : 256
       }
     }
   }
 }
```

Due to the `rating` value of 4.9, a floating-point number, the type is determined to be `float`.

As the `release_year` value is in year format (ISO format), the type is determined to be `date`.

The `title` is textual information, so a `text` data type is assigned.

The `fields` object indicates the second data type defined for the same `title` field (multiple data types).

Figure 4.2 Movie index mapping derived from document values